

NGUYEN TRAN TRUNG KIEN

AI/ML RESEARCH INTERN | MULTI-AGENT SYSTEMS | MULTIMODAL AI

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PROFILE

Final-year Artificial Intelligence undergraduate at HCMUS, expected to graduate in 2027, with research-oriented experience in multimodal model adaptation, OCR evaluation, knowledge graphs, and agentic retrieval. Built and evaluated a Vietnamese handwriting OCR fine-tuning pipeline that reduced mean CER by 68.4% relative to the base model.

EDUCATION

VNU-HCM University of Science (HCMUS)

2023–2027

B.S. in Artificial Intelligence, GPA: 3.81/4.00

Ho Chi Minh City

- Ranked 1st in the AI major by semester GPA and received the merit-based scholarship, Semester 1 AY 2023–2024; consistently among the cohort's highest-GPA students.

SELECTED RESEARCH & ENGINEERING PROJECTS

DeepSeek-OCR Fine-tuning for Vietnamese Handwriting Individual project

2025–2026

- Fine-tuned DeepSeek-OCR with 16-bit LoRA (rank 16) using Unsloth, Transformers, PEFT, FP16, gradient checkpointing, and a custom multimodal data collator on a Tesla P100.
- Trained on 5,783 UIT-HWDB line images and evaluated on 1,446 held-out samples; reduced mean CER from 38.49% to 12.15% (68.4% relative reduction) and increased exact transcription from 0.1% to 21.9%.
- Built a reproducible evaluation suite covering median/P90 CER, substitution/deletion/insertion errors, timeout handling, hallucination cases, and qualitative best/worst-case analysis.

Agentic Temporal RAG Research prototype

2026–Present

- Designed a five-stage LangGraph architecture: query planner, Neo4j temporal-graph navigator, FAISS fallback retriever, abductive evidence verifier, and abstention-aware response generator.
- Prototyped typed state transitions and fallback routing for incomplete graph evidence; investigating multi-hop temporal QA and calibrated abstention without claiming completed benchmark results.

Vietnamese Legal Knowledge Graph Team project – graph construction contribution

2026

- Developed Kaggle preprocessing and Neo4j bulk-import workflows for roughly 1.5 GB of legal data, loading 7,833,706 nodes and 19,994,844 relationships locally in about 15 minutes.
- Automated integrity checks and tuned Docker/WSL memory settings for an 8 GB machine; documented a one-command local setup for teammates.

TECHNICAL SKILLS

Research: Experimental design, ablation planning, error analysis, reproducibility, technical reporting

AI/ML: PyTorch, Transformers, PEFT/LoRA, multimodal learning, RAG, LangGraph, scikit-learn

Retrieval & Data: Neo4j/Cypher, FAISS, ChromaDB, NumPy, SQL, data pipelines

Engineering: Python, C++, FastAPI, Docker, Git/GitHub, Kaggle, Azure ML Studio

CREDENTIALS

Certifications: Deep Learning with PyTorch; Machine Learning Pipelines with Azure ML Studio; Data Visualization using Plotly.

English: IELTS Academic 6.5 (Reading 8.5, Listening 6.5, Writing 6.0, Speaking 5.5).